

Mastering Cloud Infrastructure in Google Cloud, Azure, and AWS

**Kellyn Gorman, Engineer
and Advocate
Redgate Software**



Kellyn Gorman

Multiplatform and AI Advocate/Engineer

<https://dbakevlar.com>

<https://linkedin.in.kellyngorman>

30 years experience in relational data, including Oracle, SQL Server and Postgres

DevOps with AI specialist around data protection

Former Oracle ACE Director and Microsoft Champion



Session Overview

Why multi-cloud matters

Scope: Compute, Storage, Networking

Compare AWS, Azure, GCP services

Key patterns, best practices, workload placement

The Multi-Cloud Landscape

MULTI-CLOUD ADOPTION
DRIVERS (RESILIENCE, COST,
VENDOR FLEXIBILITY)

CHALLENGES (SKILLS,
COMPLEXITY, GOVERNANCE)

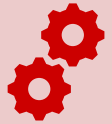
FOCUS: ALIGNING
WORKLOADS WITH THE
RIGHT CLOUD SERVICE



VM Compute: Core Concepts



vCPU / memory ratios



Instance families (general-purpose, compute-optimized, memory-optimized)



GPUs & accelerators for AI/ML workloads

Storage Models



Block Storage: VM-attached disks



Object Storage: scalable, low-cost



File/Network Storage: shared workloads



Cold/Archival tiers

AWS

AWS Compute

EC2: wide family types, Nitro System

Graviton (ARM-based) for cost/performance gains

Elastic Beanstalk, ECS, EKS for containers

Licensing options for Databases

AWS VM Instance Families

General Purpose Instances (M family, i.e. M6i, M7i)

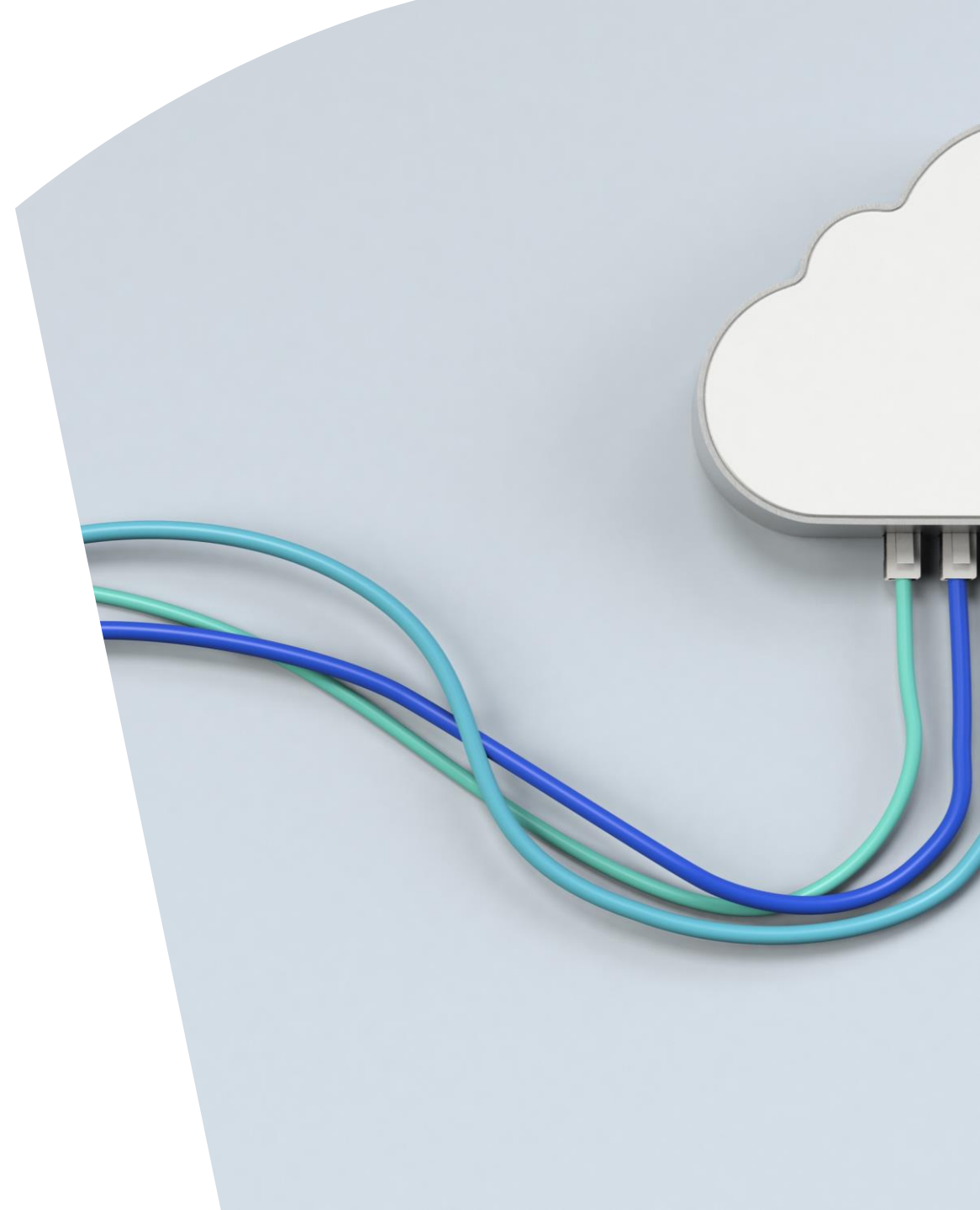
- Balanced CPU, memory and network
- Good for OLTP or dev/test
- Great for Windows licensing challenges with RDBMS
- PostgreSQL dev/test clusters do well here, too

Compute Optimized Instances (C family, i.e. C6i, C7i)

- High vCPU to memory ratio
- Best for CPU-bound databases
- Less RAM per vCPU, so not a good choice for memory-hungry Oracle or Postgres deployments

AWS Memory Optimized VM Family Instances

- R Family (R6i, R7i)
 - High memory, balanced for general DB workloads
 - Ideal for PostgreSQL and Oracle with large buffer pools or SQL Server OLTP with big caches.
- X Family (X2idn, X2iedn, X2iedn, X2iezn, X7g)
 - Extra high memory(100's of GiB up to 3+ TiB)
 - Design for in-memory DBs, including Oracle Times Ten, PostgreSQL TimescaleDB, SAP HANA)
 - May be too expensive for SQL Server licensing to use in AWS.
- Z Family (z1d)
 - High, single-thread performance with high GHz CPUs
 - Good for DBs with sensitive to per-core performance.



I and D VM Instance Families

- Designed for high IOPS and low Latency Storage
- OLTP with massive transaction rates or workloads need large local NVMe SSDs
- I Family (I4i, I3en)
 - NVMe SSDs with very high IOPS and throughput
 - Excellent for write-heavy workloads(WAL intensive Postgres, Oracle Temp Tablespace, SQL Server Tempdb)
 - Local SSDs are ephemeral – so pair with EBS for persistence!
- D Family (D3, D3en)
 - Dense HDD storage, optimize for throughput, not IOPS
 - Best for Data warehouses or log storage, not OLTP.

AWS Instance Summary

Instance Family	Known For:
M	Balanced workloads
C	CPU-bound workloads
R/X	Memory-bound workloads
I	Storage-bound, (high IOPs) workloads
D	Storage-throughput needs
Z	High GHz workloads
Graviton	Cost-optimized and Postgres only

AWS Storage



EBS: block storage for EC2



S3: object storage (11 9s durability)



EFS: elastic NFS file system



FSx: managed NetApp, Lustre, Windows FS

Storage

- EBS (Elastic Block Store)
 - Common for SQL Server and Postgres data files, logs and backups.
 - Gp3 SSD: balanced and adjustable IOPS.
 - Io2/io2 Block Express: high-endurance, very high IOPS/throughput.
 - St1/sc1: HDD, only for archival/logging, not OLTP though.
- Instance Store (local NVMe)
 - Blazing fast, but remember- ephemeral! Best for Postgres WAL, SQL Server TempDB, Oracle temp tablespace, etc.



Practical Recommendations



OLTP & Mid-sized: M6i/M7i general purpose + EBS gp3/io2



Enterprise, core-licensed or memory-heavy: R6i/X2idn with io2 Block Express



TempDB/Temp Tablespace bottlenecks: Add I4i with NVME instance to store on ephemeral for performance.



Analytics/time-series: X2iedn (lots of memory) or I4i (fast local SSDs)



Cost-efficiency: Graviton (M7g/R7g) with gp3 volumes

Azure

Azure Compute

Virtual Machines: Dv, Ev, Mv, H-series HPC

Azure VM Scale Sets for autoscaling

Azure Kubernetes Service (AKS)

Windows/SQL Server integration

Azure VM Families

General Purpose (Dsv5, Dasv5, etc.)

- Balanced CPU/Memory, good for standard OLTP workloads.
- SQL Server standard edition choice to avoid over-licensing per-core.
- Dev/test for small production Postgres

Remember letters stand for something:

- D(family)
- Default(Intel) A(AMD)
- S ephemeral SSD local disk attached
- V# is the version. Different families=different current versions.

Memory Optimized VM Family

M-series/Mv2 (Older M Family)

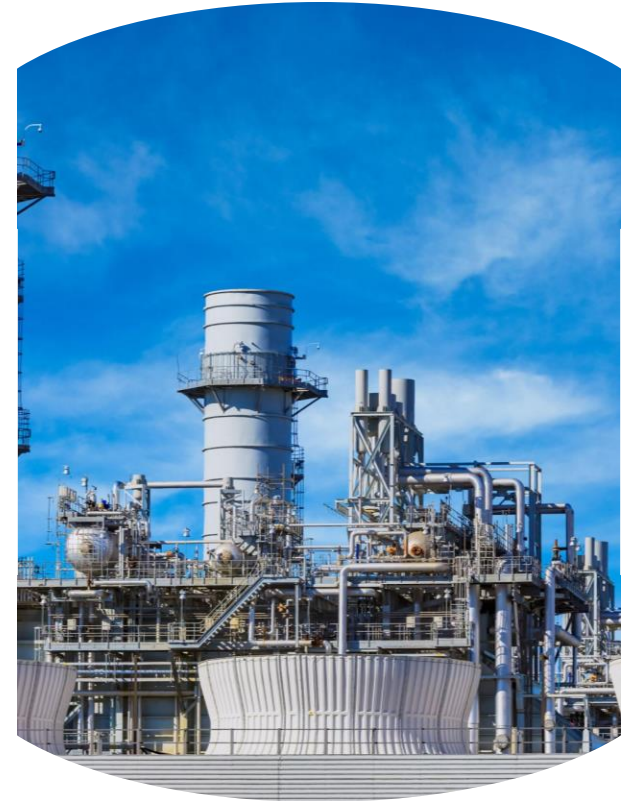
- High memory
- Low throttle values(I'll explain)
- High CPU

Mv3

- High memory and CPU
- Newest NIC(highest network speeds)
- High throttle values!

Esv5 (Ebds5/Epds5, etc.)

- Balanced CPU/memory
- Best for most enterprise, mission critical database workloads



In Summary

VM Family	Purpose
D, Smaller E Series	General OLTP Dev/Test
Esv5 Family	Enterprise, Heavy OLTP
M series	Data Warehouse/BI Workloads
L series	Some Analytics Workloads, but storage is ephemeral

Azure Storage



Azure Disks: Premium SSD, Ultra Disk



Blob Storage: Hot, Cool, Archive tiers



Azure Files: SMB/NFS shared storage



Azure NetApp Files: enterprise file storage

Storage

VM decides how much IO you receive!!!*

Premium SSD v2

- Configurable IOPS & Throughput per disk
- Up to 80K IOPS and 1200 MBPs throughput*
- First choice for database workloads

Ultra Disk (Go-to for Azure, but should it be?)

- .5 ms latency and up to 160K IOPs and 4000 MBPs throughput*
- Perfect for SQL Server OLTP/high concurrency and write-heavy Postgres
- Can be costly

Not for Database Workloads

Standard SSD

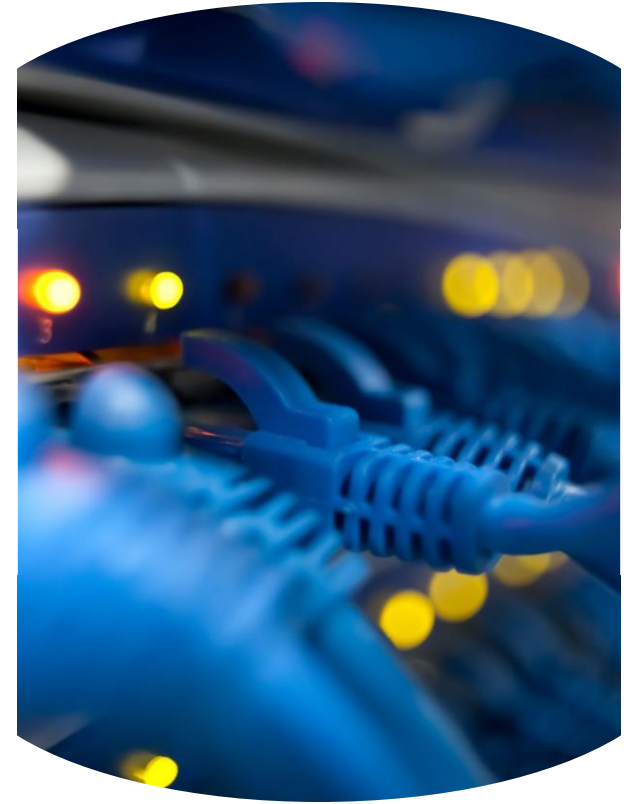
- Lower IOPS (500-6000)
- OK for dev/test or low-traffic Postgres
- Not recommended for production.

Standard HDD

- Just a hard NO for any datafiles for any platform.
- Suitable for backups, archives or play environments.

Network Attached Storage

- Only Limited by the NIC, not the VM Attached Storage Limit
 - Traditionally much higher IOPs and throughput.
- Azure NetApp Files
- Silk
- Elastic SAN
- Pure Storage for Cloud



Recommended Storage Practices



Separate datafiles from conflicting IO demands

Transaction/archival logs
ETL load files
Installation or database engine



Tempdb/temp tablespace- consider using ephemeral local disk with automated recreation before startup if required.



Enable read caching, (host level caching) on datafiles, but disable any write caching on WAL/logging, etc.



Use Availability sets and Availability zones for HA and use managed disk

Google Cloud

Google Cloud Compute

Compute Engine: E2, N2, C2, Tau T2D ARM

Preemptible VMs for cost savings

Google Kubernetes Engine (GKE)

AI/ML GPU & TPU offerings

Google Cloud Compute Instance Family

- General Purpose (N2, N2D, E2 C3) Standard
 - N2 is Intel(Cascade/Ice Lake) Good for OLTP and mid-size Postgres
 - N2D is AMD EPYC, more cost effective, (about 20% cheaper per core)
Great for Postgres dev/prod where licensing isn't core-bound.
 - E2 is a great budget option and has lower performance consistency.
Good for dev/test.
 - C3 Standard: Latest generation with Intel Sapphire Rapids chip.
Balance but with newer networking and storage bandwidth.

Compute-Optimized (C2, C3)



High GHz per core.



Best for CPU-intensive workloads.

Procedure-heavy SQL Server

Highly transactional Oracle

Heavy analytics joins and aggregates in Postgres



Lower memory per vCPU

Not ideal for memory-heavy Oracle or Postgres caching workloads.

Memory-Optimized (M1, M2, M3)



Massive Memory per VM, (up to 12TB!)



Suited for:

SQL Server enterprise workloads with multi-TB buffer pools
Large analytics, time-series or in-memory extension Postgres



Costly, often used in regulated environments, such as financial services, SAP HANA, etc.)



Memory and CPU demanding Oracle workloads, with less demanding IO

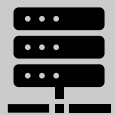
Specialty VMs



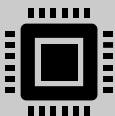
Tau (T2D, T2A, C3A) Optimized for price/performance

T2D is AMD EPYC Milan chip and priced well for Postgres OLTP

T2A is ARM-based. Great for Postgres (native support)



Sole-Tenant Nodes: Useful for Oracle and SQL Server licensing (per core/socket) and isolation requirements.



Bare Metal Solutions: Great for very large SQL Server deployments with specific hardware/OS requirements beyond standard VMs.

In Summary

VM Instance Family	Purpose
N2/N2D	Small- medium OLTP
M2/M3	Enterprise OLTP medium- large
Licensing Optimization	Sole-tenant nodes
T2D	Extreme OLTP
M-series	Analytics or Time-series workloads
E2 with pd-standard	Cost-sensitive dev/test

Google Cloud Storage



Persistent Disk: block storage for VMs



Cloud Storage (GCS): object storage tiers



Filestore: managed file storage (NFS)



Cloud Volumes (NetApp/partners)

Persistent Disks

Balanced PD (pd-balanced)

- Cost-effective SSD with predictable performance
- Good for general Postgres/SQL Server Workloads

Extreme PD (pd-extreme)

- High IOPS (up to 120K per disk)
- VM still decides how much IO you can achieve.)
- Equivalent to Azure Ultra Disk or AWS io2 Block Express)

Standard PD (pd-standard)

- HDD-backed. Suitable for backups and archives, etc. Not for production OLTP/workloads.

Local SSD

- Attached physically to VM host.
- Ultra-low latency, high IOPS (~2M per VM)
- Data is EPHEMERAL, secondary storage solution is important.
- Great for:
 - SQL Server TempDB
 - Oracle Temp Tablespace
 - Postgres WAL logs or temp space.



Filestore



NFS Managed Storage/Managed file shares



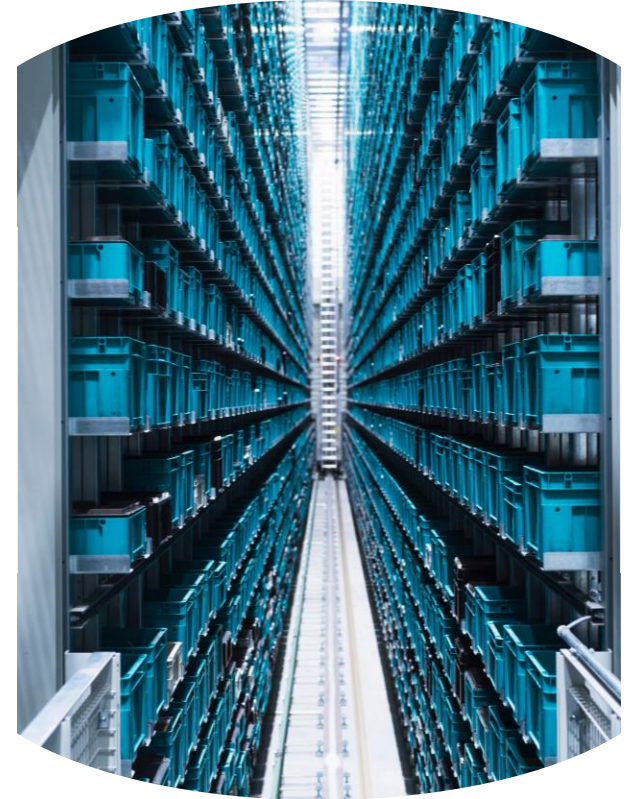
More commonly used for app data than datafiles.



Can be used for backups, staging or read replicas

Network Attached Storage

- Yes, it's a thing in GCP, too!
- Only limited by the NIC, not the VM attached storage limit.
- Numerous Solutions
 - Google NetApp Files
 - Silk
 - Pure Storage



Recommended Practices

1

Separate Disks for
data, logs and
tempdb/WAL

2

Be aware of
limits/quotas on VM
specific, not just
family

3

Stripe multiple PDs
with RAID0(mdadm
or LVM) to increase
throughput/IOPS on
standard disks

4

Remember that
local disks are
ephemeral.

AWS vs. Azure vs. GCP Recap

Workload	AWS	Azure	GCP
Balanced OLTP	M6i/M7i + gp3	Dv5/Dsv5 + Premium SSD v2	N2/N2D + pd-balanced
CPU-bound	C6i/C7i	Fsv2	C2/C3 High-CPU
Memory-heavy	R6i/X2idn	Ev5/M-series	M2/M3
Storage-heavy (tempdb/WAL)	I4i (NVMe)	Lsv3 (NVMe)	Local SSD(NVMe)
Extreme IOPS	io2 Block Express	Network Attached Storage	pd-extreme or Network Attached
Cost-optimized	Graviton (R7g/M7g)	Dasv5 (AMD)	T2D (AMD), T2A (ARM for Postgres only)

Networking Fundamentals

VIRTUAL NETWORKS: VPC
(AWS), VNET (AZURE), VPC (GCP)

SUBNETS, FIREWALLS, ROUTING

BANDWIDTH AND LATENCY
CONSIDERATIONS

AWS Networking

VPC + TRANSIT GATEWAY

DIRECT CONNECT FOR
HYBRID CLOUD

LOAD BALANCERS: ALB, NLB,
GWLB

STRONG GLOBAL REACH

Azure Networking

VNet & Peering

ExpressRoute for hybrid
connectivity

Load Balancers: Azure LB,
App Gateway, Front Door

Deep Microsoft integration

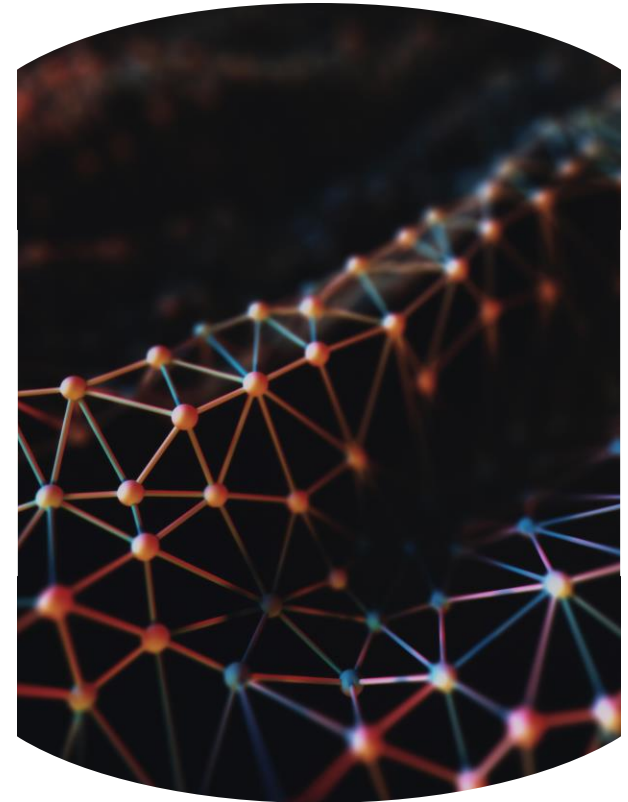
Google Cloud Networking

Global VPC (unique to GCP)

Cloud Interconnect & VPN

Load Balancers: Global HTTP(S),
TCP/UDP, Internal LB

High-performance backbone



Data Services

AWS Cloud Services

Data Services



Managed by AWS, not just the infrastructure.

This is PaaS and SaaS options that are most likely connected to database workloads.

AWS Data Migration and Integration

- Data Migration Service (DMS)
 - Move data in and out of EC2-hosted databases.
 - Supports homogenous and heterogenous migrations.
- AWS Data Sync
 - Move files into a staging area for bulk loads
- AWS Schema Conversion Tool (SCT)
 - Converts schemas from one platform to another.
 - Handles most common migration scenarios.
 - Migrates 60% of code, not just objects.

Application and Analytics Services

- AWS Lambda
 - Connect using RDS proxy, Secrets Manager or directly via VPC.
- Amazon API Gateway
 - Front-end API integration with backend database solution.
- Amazon AppFlow
 - SaaS integrations that can push data into databases.
- Amazon Athena Federated Query
 - Query databases via connectors.
- Redshift Spectrum
 - Can query databases as external data sources.
- SageMaker
 - Pull data directly from a database.
- AWS Glue
 - ETL engine that connects via JDBC/ODBC

Monitoring, Secrets and Ops

- Amazon CloudWatch/Logs
 - Monitor DB metrics, integrate with custom exporters
- Systems Manager
 - Patching and automation for EC2 hosted DBs
- AWS Backup
 - Centralized Backup for EBS volumes hosting a database.
- Amazon VPC, Transit Gateway, Direct Connect
 - Secure networking for hybrid cloud database environments
- AWS PrivateLink
 - Private endpoint access
- Secrets Manager
 - Manage database credentials

PaaS Database Options

PostgreSQL

- Amazon RDS for PostgreSQL
 - Vanilla Postgres (community)
 - Managed backups, patching and replication.
- Amazon Aurora
 - AWS's cloud-native Postgres compatible
 - Faster scaling, replication and distributed storage

SQL Server

- Amazon RDS for SQL Server
 - Managed SQL Server, Standard, Enterprise, Web and Express
 - Limited licensing rules
 - OS and patching managed by AWS
- Amazon RDS for Oracle
- Managed Oracle standard and enterprise edition.
- Options for HA with RAC. May be at risk with Exadata@cloud integration.

Azure Cloud Services

Data Integration and Migration

Azure Data Factory

- ETL/ELT pipelines with connectors for most database platforms.

Database Migration Service (DMS)

- Purpose-built for migrating numerous db platforms to Azure, either VMs or PaaS.

Azure Data Share

- For Distributing DB extracts to other Azure tenants.

Analytics and AI

All about Fabric!

- Synapse Analytics (SQL Data Warehouse)
 - Can use connectors, mirrors or ADF, even linked services to pull from other databases.
- Azure Machine Learning
 - Connects directly to databases for training ML models
- Power BI
 - Native connectors for many databases and common for reporting by experts and novices alike.
- Azure Databricks
 - Spark-based platform that integrates with most databases for analytics/ML

Monitoring, Secrets & Ops

- Monitor/Log Analytics
 - Collect custom DB metrics with agents or exporters
- Azure Arc-enabled databases
 - Hybrid managed/monitoring of IaaS databases
- Azure Automation/Update Mgmt
 - OS and patch scheduling for VMs
- Azure VNet, ExpressRoute, VPN Gateway
 - Secure connectivity for hybrid or cross-service DB access.
- Private Link
 - Private endpoints for databases running on VMs.
- Azure Key Vault
 - Store DB credentials, integrate with apps.

PaaS Database Services

- Azure SQL Database
 - Fully managed, single databases or elastic pools
 - Feature subset of SQL Server
- SQL Managed Instance
 - PaaS, but with close-to full SQL compatibility.
 - Ideal for lift and shift options.
- SQL Server on Azure VMs (images)
 - Not really PaaS, but....
- Azure Database for PostgreSQL
 - Managed Postgres service
 - Supports custom configs, HA and scaling.
- Azure DB for PostgreSQL- Hyperscale Citus
 - Sharded, horizontally scalable Postgres
 - Best for multi-tenant SaaS or large-scale.
 - Azure Arc-enabled PostgreSQL
 - Managed Postgres via Kubernetes
 - Can run on infra or hybrid cloud.

Google Cloud Services

Data Integration & Migration

- Data Migration Service (DMS)
 - Fully managed service to migration databases into Cloud SQL.
 - May use partner tools like Striim, Attunity, etc.)
- Dataflow
 - Streaming/batch ETL. Uses JDBC/ODBC to connect to many databases.
- Dataproc (managed by Hadoop/Spark)
 - Pulls data from databases for big data processing.
- BigQuery Data Transfer Service
 - Partner connectors(Fivetran, Informatica) can move databases data into BigQuery

Analytics and AI

Big Query

- Core analytic service. Connects to most databases via federated queries (via BigQuery Omni or external connectors)

Looker, Looker Studio, Data Studio

- Native BI Connectors for databases.

Vertex AI

- ML pipelines can read directly from databases via JDBC/ODBC or use Dataflow as staging.

Monitoring, Secrets and Ops

- Cloud Monitoring/Cloud Logging
 - Collect custom metrics/logs from databases
- Config Connector & Deployment Mgr
 - IaC, including DB VM provisioning.
- Backup and DR Service
 - Backup policies can be applied to persistent disks for hosted DBs
- VPC, Cloud VPN, Interconnect
 - Secure hybrid connectivity
- Private Service Connect
 - Private endpoints into GCP-hosted databases on MVs.
- Identity-Aware Proxy(IAP)
 - Secure DB Admin access without exposing public Ips.
- Secret Manager
 - Store and rotate DB credentials

PaaS Database Services

PostgreSQL

- Cloud SQL for PostgreSQL
 - Managed PostgreSQL(patching, backups, HA)
 - Standard community version with limited extensions.)
- AlloyDB/AlloyDB Omni
 - Cloud native Postgres-compatible
 - 4x faster compared to Cloud SQL
 - Competes with AWS Auroral and Azure Hyperscale
 - Omni can be decoupled from the cloud and run anywhere.

SQL Server

- Cloud SQL for SQL Server
 - Managed SQL Server(Express, standard Enterprise)
 - Supports backups, HA, AD Authentication
 - Limited feature set.

Oracle has Exadata@Cloud, too!

AWS vs. Azure vs. Google Cloud Services

Capability	AWS	Azure
VM-hosted DB integration	DMS, Glue, Athena Federated, Secrets Manager, CloudWatch	Data Factory, DMS, Synapse, Key Vault, Monitor
Managed SQL Server (subset)	RDS for SQL Server	Azure SQL Database
Managed SQL Server (full compatibility)	NONE	SQL Managed Instance
Managed PostgreSQL	RDS Postgres	PostgreSQL Flexible Server
Cloud-native Postgres	Aurora PostgreSQL	PostgreSQL Hyperscale (Citus)
Extreme analytics	Redshift	Synapse Analytics

Architectural Patterns

Lift & shift vs cloud-native

Hybrid/multi-cloud
placement

Resiliency patterns (multi-
region, multi-zone)

Cost optimization levers



Best Practices Across Clouds



Standardize monitoring/logging



Use IaC (Terraform, Pulumi, Bicep)



Governance: tagging, cost mgmt, security baselines



Plan for data gravity & egress costs

Key Takeaways



AWS = breadth & maturity



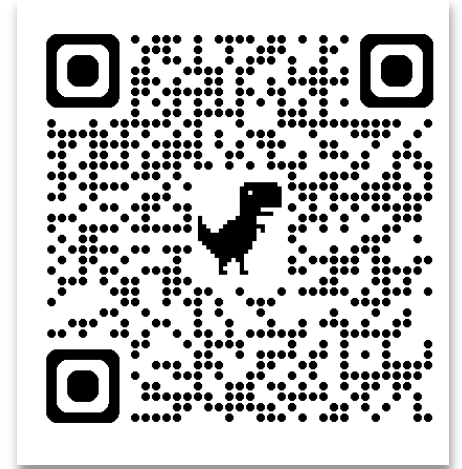
Azure = enterprise integration and AI



GCP = analytics & innovation



Each cloud has an offering combination to meet demands, but also strength in specific areas.



Thank you for attending

Kellyn.Gorman@red-gate.com